

CM5 MINIMA Carrier Board

Technical Reference Manual

Document Version:	1.0
Hardware Revision:	V1
Date:	2025-12-04
Organization:	PSYC Aerospace and Defence Industries Pvt. Ltd.
EDA Tool:	KiCad E.D.A. 10.0.3

This document describes the electrical architecture, power system, communication interfaces, onboard sensors, external I/O routing, and schematic references for the CM5 MINIMA Raspberry Pi Compute Module 5 carrier board.

Confidential Engineering Reference

PSYC Aerospace and Defence Industries Pvt. Ltd.

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1 System Overview

The **CM5 MINIMA** is a compact and highly integrated carrier board designed for the **Raspberry Pi Compute Module 5**. The board is intended for aerospace, telemetry, embedded autonomy, advanced flight control support, data logging, and edge-compute applications.

The design breaks out mission-critical interfaces from the Compute Module 5, including:

- USB Type-C power input.
- PCIe M.2 expansion slot for NVMe storage or compatible PCIe modules.
- Gigabit Ethernet interface with magnetics, common-mode filtering, and ESD protection.
- USB 2.0 hub subsystem for downstream peripheral expansion.
- MIPI DSI/CSI FPC connector.
- Onboard flight sensors over I2C.
- External I/O headers for flight-controller, telemetry, CRSF, LED, SD card, and GPIO routing.

Design Intent: The board is optimized for compact embedded aerospace systems where the Compute Module 5 acts as the onboard companion computer, sensor processor, logger, telemetry interface, or autonomy processor.

2 Key Specifications

Parameter	Description
Compute Module	Raspberry Pi Compute Module 5
Main Input Power	+5V through USB Type-C power connector
System Logic Rail	+3V3_PI
PCIe Expansion	M.2 socket with dedicated 3.3V rail
USB Expansion	USB2514B USB 2.0 hub controller
Ethernet	CM5 Ethernet breakout with external magnetics, common-mode chokes, and ESD protection
Onboard IMU	ICM-45686 6-axis inertial sensor
Barometric Sensor	BMP585 pressure sensor
Sensor Bus	I2C1 bus after V1 hardware patch
EDA Tool	KiCad E.D.A. 10.0.3
Schematic Revision	V1
Schematic Date	2025-12-04

3 Power Architecture

The CM5 MINIMA board is powered from a standard USB Type-C power input. The main input rail is distributed as a +5V rail across the board and is used as the upstream supply for local subsystems.

3.1 Main Input

Item	Implementation
Connector	USB Type-C power connector, TYPEC-305-ACP16H458
Input Rail	+5V

Primary Use	CM5 power input, USB subsystem, and downstream regulator input
Protection	Type-C input ESD protection using RCLAMP0502BA on USB data lines

3.2 System Logic Rail

The board uses the +3V3_P1 rail for logic-level subsystems, GPIO routing, I2C sensors, and support circuitry.

All onboard logic devices connected to CM5 GPIOs should remain compatible with the CM5's 3.3V logic domain.

3.3 PCIe M.2 Power Delivery

The M.2 socket uses a dedicated high-efficiency buck converter to generate a stable 3.3V supply rail.

Item	Implementation
Regulator	AP3441SHE-7B buck converter
Input	+5V
Output	M2_3V3
Load	M.2 PCIe socket, typically NVMe SSD or compatible expansion device
Output Filtering	Local ceramic capacitors and bulk capacitance near M.2 power rail

Design Check: Ensure the M.2 3.3V regulator current rating is sufficient for the selected NVMe drive. Some NVMe SSDs can demand high peak current during startup and write bursts.

4 Communication and Interface Subsystems

4.1 USB 2.0 Hub Subsystem

The CM5 native USB 2.0 interface is expanded using a Microchip USB hub controller.

Function	Implementation
Hub Controller	Microchip USB2514B USB 2.0 hub controller
Crystal	24.000 MHz crystal, 7M-24.000MEEQ-T
Upstream Interface	CM5 USB 2.0 data pair
Downstream Ports	Multiple USB 2.0 downstream channels
Power Switching	MIC2026-1YM dual-channel power distribution switch
Additional Current Limit	AP22653W6 switch
ESD Protection	RCLAMP0502BA on Type-C USB data lines

4.1.1 USB Power Distribution

USB downstream power is switched and current-limited to protect the carrier board and attached peripherals. The current-limit switches isolate excessive load conditions from the rest of the board.

For field deployment, USB peripherals should be verified under cold boot, hot plug, and brownout conditions.

4.2 Ethernet Interface

The Ethernet front-end is designed for improved robustness in electrically noisy environments such as UAVs, robotics platforms, and motor-driven systems.

Function	Implementation
Ethernet Source	CM5 Ethernet differential pairs
Magnetics	Würth Elektronik WE-STST_4532 transformers
Common Mode Filtering	Würth WE-CNSW_0603 wire-wound common-mode chokes on Ethernet pairs
Protection	TPD4E05U06 / TPD4EUSB30 ESD protection arrays
Connector	External Ethernet connector header/interface

4.2.1 Ethernet Signal Chain

The intended Ethernet signal path is:

CM5 Ethernet Pairs → Magnetics → Common Mode Chokes → ESD Protection → Ethernet Connector

Layout Note: Ethernet differential pairs must be routed with controlled impedance, length matching, and continuous return paths. Avoid routing Ethernet pairs close to high-current switching nodes, buck inductors, motor wiring, or RF transmitters.

4.3 PCIe M.2 Interface

The board exposes the CM5 PCIe lane to a standard M.2 socket.

Function	Implementation
Connector	M.2 socket, J701
Interface	PCIe differential transmit and receive pairs
Power Rail	M2_3V3 generated from AP3441SHE-7B buck converter
Reference Clock Component	ASEK-32.768KHZ-L-R-T oscillator in schematic section
Mechanical Support	M.2 solder nut and screw mounting
Activity Indicator	M.2 activity LED line

PCIe Layout Requirement: Route PCIe TX/RX differential pairs as high-speed controlled-impedance lines. Keep stubs short, avoid unnecessary vias, and maintain pair-to-pair spacing to reduce crosstalk.

4.4 MIPI DSI / CSI Connector

A 22-pin FPC connector provides access to the CM5 MIPI D-PHY interface.

Function	Implementation
Connector	22-pin FPC connector, J802
Interface	DPHYO_C MIPI D-PHY lanes
Auxiliary Lines	CAM_GPIO0, CAM_GPIO1, SDAD, SCLD
Power Reference	+3V3_PI present for auxiliary interface support
Ground	Dedicated ground pins and return path

5 Onboard Flight Sensors

The CM5 MINIMA includes onboard flight-oriented sensors connected to the I2C bus. These sensors are powered from the clean +3V3.PI rail.

Component	Function	Description
ICM-45686	6-Axis IMU	Provides gyroscope and accelerometer measurements for attitude estimation, motion sensing, and stabilization support.
BMP585	Barometric Sensor	High-precision pressure sensor used for altitude estimation, telemetry, and vertical trend measurement.

5.1 Sensor Bus Migration

In the V1 hardware patch, the onboard sensor data lines were migrated from the MIPI-related bus to the standard I2C1 bus for improved software compatibility and easier Linux driver access.

Hardware Patch 1: BMP585 and ICM-45686 sensor data lines are routed to I2C1 instead of the previous MIPID bus mapping.

5.2 I2C Pullups

The sensor section includes pull-up resistors on the I2C lines. These maintain valid logic levels for the I2C bus and allow open-drain communication between the CM5 and the onboard sensors.

Bring-Up Check: During first boot, scan the I2C bus using Linux tools such as `i2cdetect`. Confirm that both the BMP585 and ICM-45686 respond at the expected addresses before enabling flight software.

6 External I/O and Flight Controller Routing

The board provides external routing headers for flight-controller integration, telemetry, CRSF receiver connection, SD card routing, LEDs, and general-purpose signals.

6.1 Flight Controller UART

The carrier includes UART routing intended for flight controller communication. This allows the CM5 to act as a companion computer for MAVLink, telemetry processing, mission control, sensor fusion, or onboard autonomy tasks.

Interface	Purpose
Flight Controller UART	Serial communication with Pixhawk-class or other autopilot controllers
Telemetry UART	External telemetry or companion radio interface
CRSF UART	Crossfire receiver or related serial RC protocol interface
GPIO Lines	Custom control, enable, status, or payload interface signals

6.2 SD Card Interface

The board includes MicroSD card routing for storage, data logging, configuration, or boot-related use depending on system configuration.

6.3 LED Indicators

Status indicator lines include power and activity indication.

Signal	Purpose
LED_ACTIVITY	System or storage activity indication
Power LED	Board power status indication
Custom LED Lines	Application-specific telemetry or debug indication

7 Recommended Software Bring-Up Procedure

7.1 Initial Power-On

1. Inspect the assembled board for shorts, solder bridges, reversed components, and connector damage.
2. Power the board using a current-limited 5V bench supply before connecting sensitive peripherals.
3. Confirm the +5V input rail.
4. Confirm the +3V3_PI rail.
5. Confirm the M2_3V3 rail before installing an NVMe drive.

7.2 Linux Peripheral Verification

After booting the CM5, verify interfaces in the following order:

1. Confirm USB enumeration using `lsusb`.
2. Confirm Ethernet link using `ip link` or `ethtool`.
3. Confirm I2C bus availability using `i2cdetect`.
4. Confirm PCIe enumeration using `lspci`.
5. Confirm NVMe storage using `lsblk`.
6. Confirm UART ports using `dmesg`, `raspi-gpio`, or application-level serial testing.

7.3 Example Diagnostic Commands

USB devices

```
lsusb
```

PCIe devices

```
lspci
```

Block devices / NVMe

```
lsblk
```

Network interfaces

```
ip link
```

```
ethtool eth0
```

I2C bus scan

```
sudo i2cdetect -y 1
```

```
# Kernel messages
dmesg | grep -i usb
dmesg | grep -i pcie
dmesg | grep -i i2c
```

8 Electrical Design Notes

8.1 Power Integrity

- Keep the +5V input path low impedance.
- Use adequate bulk capacitance near the Type-C input.
- Keep the M.2 buck converter close to the M.2 socket.
- Place high-frequency decoupling capacitors near IC power pins.
- Ensure the ground plane is continuous under high-speed interfaces.

8.2 Signal Integrity

- Route USB differential pairs as matched impedance pairs.
- Route PCIe differential pairs with controlled impedance and strict length matching.
- Route Ethernet differential pairs with symmetry and proper spacing.
- Avoid running high-speed lines under switching regulator inductors.
- Keep I2C pullups close to the bus and avoid excessive bus length.

8.3 EMI and UAV Deployment Considerations

For drone and robotics use, the board should be isolated from high-current motor wiring and switching ESC power paths. Ethernet, USB, and sensor lines should be physically separated from noisy power electronics where possible.

Flight System Warning: This carrier board should not be treated as a certified flight controller by itself. For safety-critical aircraft operation, use an independent autopilot or certified flight controller and treat the CM5 MINIMA as a companion compute board unless the full system has been qualified.

9 Connector and Subsystem Summary

Reference	Subsystem	Description
J102 / J601	USB Type-C	Main USB Type-C power and USB connection depending on schematic section
J701	PCIe M.2	M.2 expansion socket for NVMe or compatible PCIe device
J802	DSI/CSI	22-pin FPC connector for MIPI D-PHY camera/display interface
J301	Ethernet	Ethernet interface connector after magnetics and filtering
J509	MicroSD	SD card connector for storage/logging interface

U404	IMU	ICM-45686 6-axis motion sensor
A401	Barometer	BMP585 pressure sensor
U603	USB Hub	USB2514B USB hub controller
U702	M.2 Regulator	AP3441SHE-7B buck regulator for M2_3V3 rail

10 Errata and Revision History

Revision	Date	Description
V1	2025-12-04	Initial schematic release.
Hardware Patch 1	2025-12-04	Sensor data lines for BMP585 and ICM-45686 were moved from MIPID routing to the standard I2C1 bus for improved compatibility.

11 Manufacturing and Assembly Notes

- Confirm all Type-C connector pins are soldered correctly and mechanically reinforced.
- Verify orientation of ESD arrays, current-limit switches, regulators, and crystals.
- Inspect the M.2 connector solder joints under magnification.
- Verify Ethernet magnetics and common-mode choke orientation.
- Confirm I2C pullups are populated according to the final bus plan.
- Confirm all test points are accessible before enclosure integration.
- Perform continuity checks on +5V, +3V3_PI, M2_3V3, and GND before first power-up.

12 Board Bring-Up Test Checklist

No.	Test Item	Status
1	Visual inspection completed	Pass / Fail
2	No short between +5V and GND	Pass / Fail
3	No short between +3V3_PI and GND	Pass / Fail
4	+5V input rail verified	Pass / Fail
5	+3V3_PI rail verified	Pass / Fail
6	M2_3V3 rail verified	Pass / Fail
7	CM5 boots successfully	Pass / Fail
8	USB hub enumerates	Pass / Fail
9	Ethernet link detected	Pass / Fail
10	I2C sensors detected	Pass / Fail
11	PCIe device detected	Pass / Fail
12	NVMe device detected if installed	Pass / Fail
13	UART telemetry tested	Pass / Fail
14	CRSF UART tested	Pass / Fail
15	SD card interface tested	Pass / Fail
16	LED indicators verified	Pass / Fail

A Schematic Sheet Reference

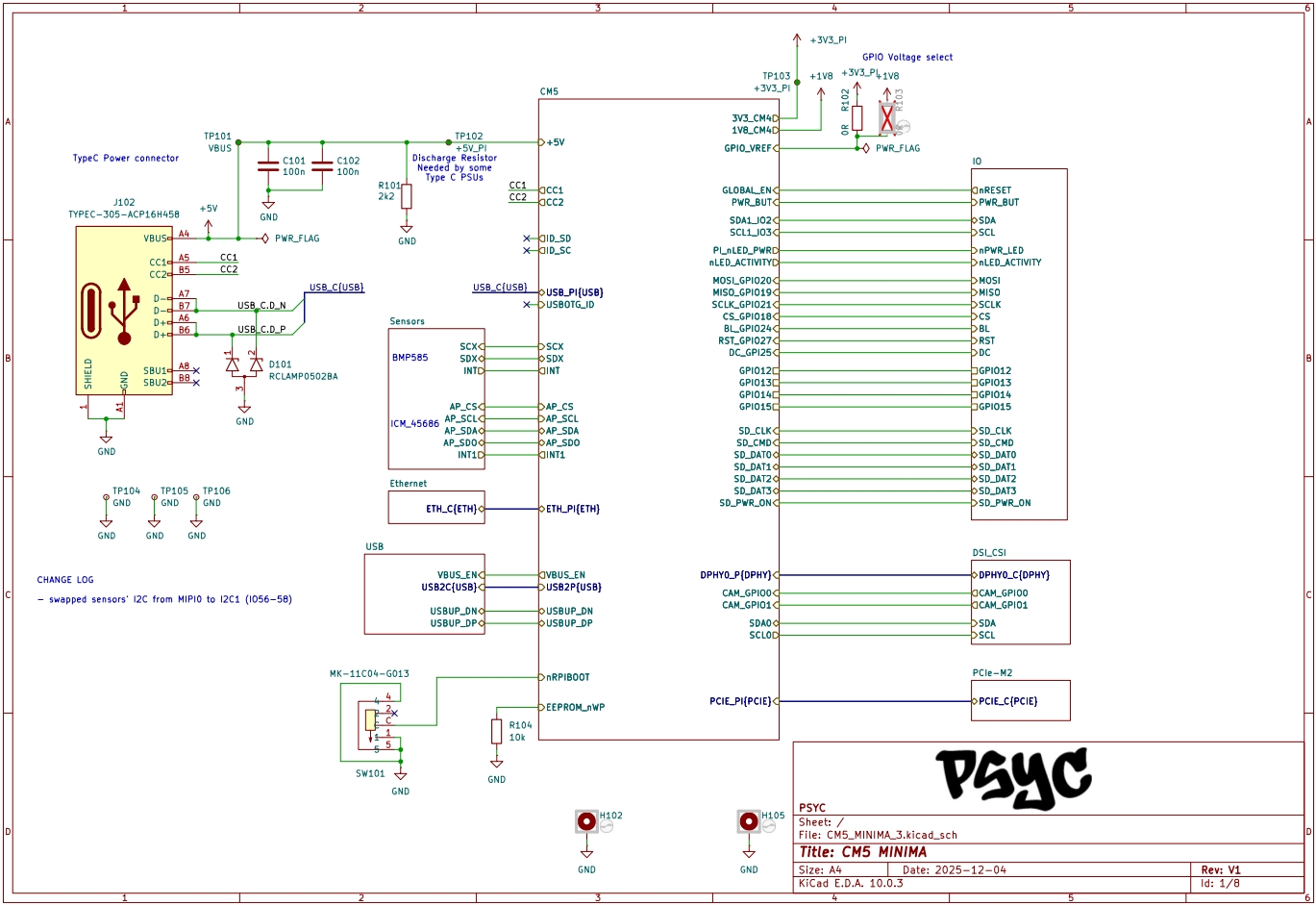
The CM5 MINIMA schematic set is organized into eight sheets:

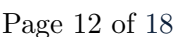
Sheet	Title	Contents
1 / 8	CM5 MINIMA Top Sheet	System-level interconnects, Type-C power input, sensor block, Ethernet, USB, DSI/CSI, PCIe-M.2 hierarchy links, GPIO routing.
2 / 8	CM5	Compute Module 5 connector pin breakout, GPIO, USB, PCIe, Ethernet, SD, and MIPI routing.
3 / 8	Ethernet	Ethernet magnetics, common-mode chokes, ESD arrays, and connector routing.
4 / 8	Sensors	ICM-45686 IMU, BMP585 barometer, I2C pullups, decoupling, and interrupt lines.
5 / 8	IOs	External I/O headers, SD card, UART/telemetry/CRSF routing, LED indicators, and support circuits.
6 / 8	USB2.0	USB Type-C interface, USB2514B hub controller, 24 MHz crystal, current-limit switching, and ESD protection.
7 / 8	PCIe-M.2	M.2 socket, 3.3V regulator, PCIe routing, activity LED, and mounting hardware.
8 / 8	DSL_CSI	22-pin MIPI DSI/CSI FPC connector and auxiliary camera/display control lines.

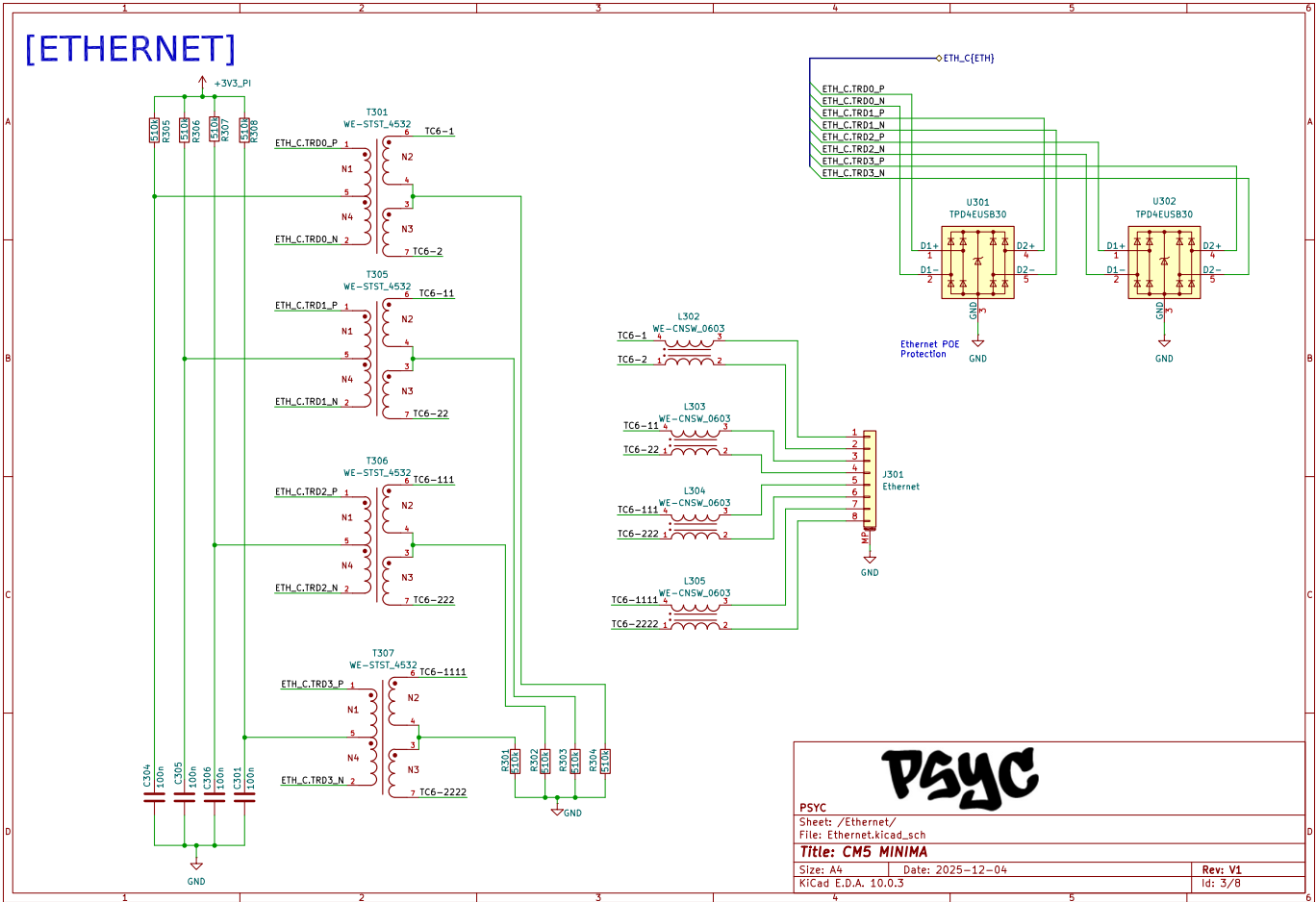
B Complete Schematic PDF

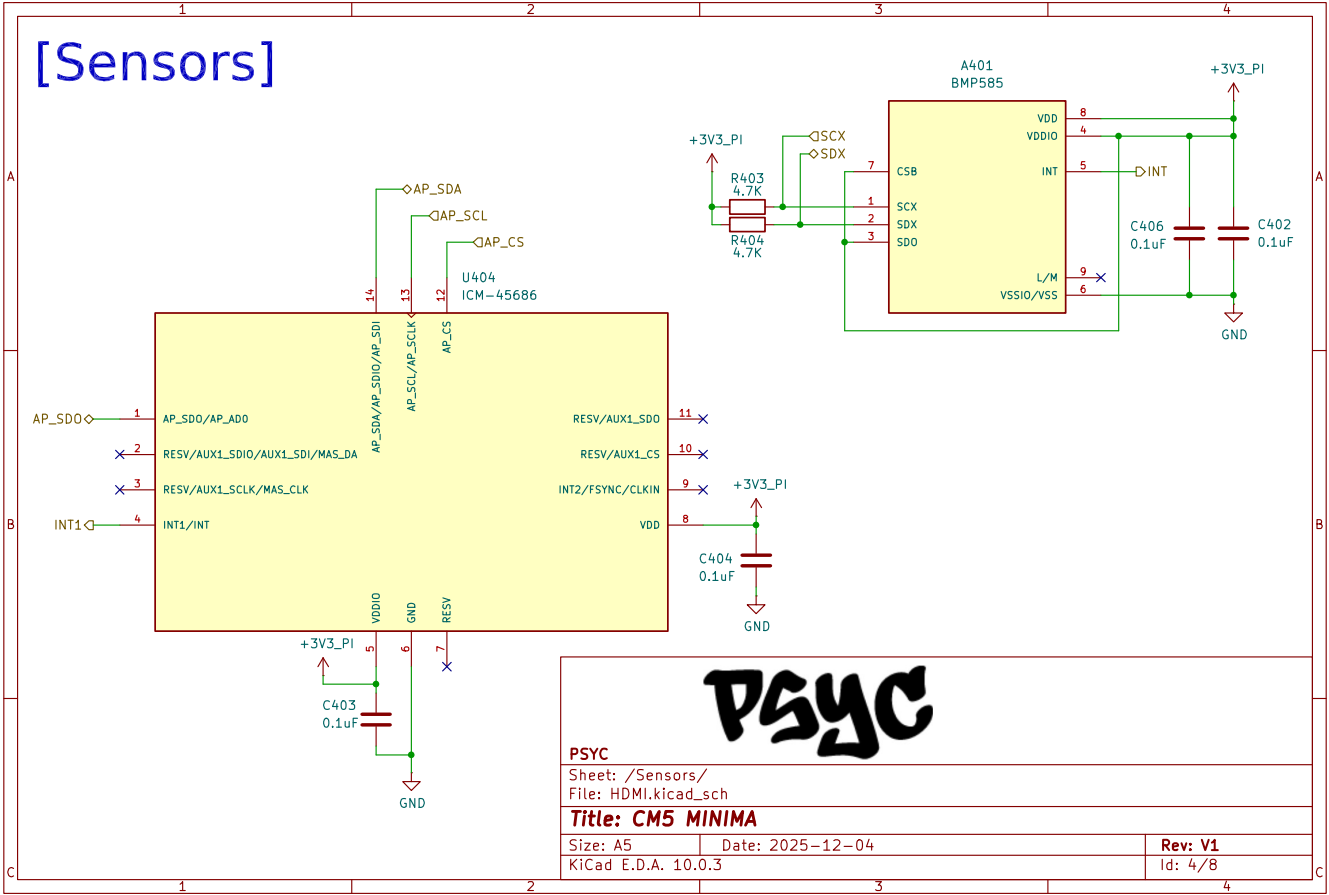
The following pages include the complete schematic reference exported from KiCad.

To compile this section in Overleaf, upload the file named `schematic.pdf` into the same project folder as this `main.tex` file.

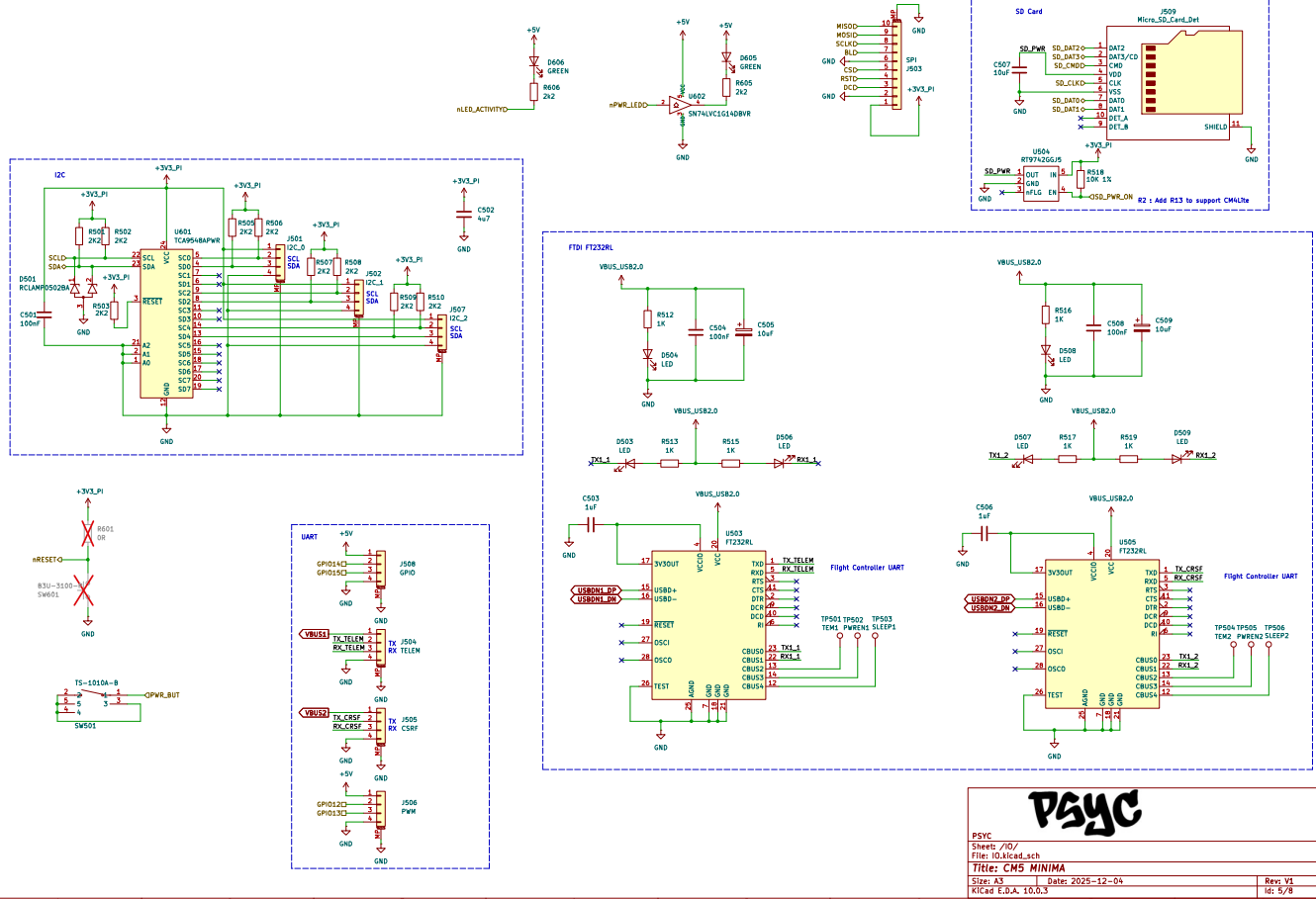








[IOs]



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